

Week 3: Seller and Inventory Management System

Scenario: The e-commerce company sells products supplied by multiple sellers. The system should maintain seller details, track products supplied by each seller, and monitor inventory levels to ensure product availability.

1. Create Seller and Inventory Tables

```
SQL> CREATE TABLE Seller(Seller_ID VARCHAR(7), Seller_Name VARCHAR(25),Email VARCHAR(30), Phone number(10), Address VARCHAR(45))
Table created.

SQL> CREATE TABLE Inventory(Inventory_ID VARCHAR(7),
2 Product_ID VARCHAR(7), Seller_ID VARCHAR(7),
3 Stock_Quantity number, Stock_status VARCHAR(11),
4 Last_Updated DATE);
Table created.
```

2. Establish Relationships

Seller and Product

One seller can supply multiple products.

One product can be supplied by one or multiple sellers.

Relationship: Seller → Product is one to many

Product and Inventory

Each product has inventory information.

Inventory tracks available stock.

Relationship: Product → Inventory is one to one Primary Key added for Both Tables:

PRIMARY KEY:

```
SQL> ALTER TABLE Seller ADD CONSTRAINT PK_Seller PRIMARY KEY(Seller_ID);
Table altered.

SQL> ALTER TABLE Inventory ADD CONSTRAINT PK_Inv PRIMARY KEY(Inventory_ID);
Table altered.
```

Foreign Key for Inventory Table:

```
SQL> ALTER TABLE Inventory ADD
2 CONSTRAINT FK_Prod FOREIGN KEY(Product_ID) REFERENCES Product(Product_ID);
Table altered.
```

```
SQL> ALTER TABLE Inventory ADD
2 CONSTRAINT FK_Seller FOREIGN KEY(Seller_ID) REFERENCES Seller(Seller_ID);
Table altered.
```

To make Email UNIQUE:

```
SQL> ALTER TABLE Seller ADD
2 CONSTRAINT Un_mail UNIQUE(Email);
```

```
SQL> INSERT INTO Seller Values ('&Seller_id','&Seller_Name','&email','&Phone','&Address');
Enter value for seller_id: S001
Enter value for seller_name: Arjun
Enter value for email: arjun@gmail.com
Enter value for phone: 9876543212
Enter value for address: Salem
old 1: INSERT INTO Seller Values ('&Seller_id','&Seller_Name','&email','&Phone','&Address')
new 1: INSERT INTO Seller Values ('S001','Arjun','arjun@gmail.com',9876543212,'Salem')

1 row created.

SQL> /
Enter value for seller_id: S002
Enter value for seller_name: Priya
Enter value for email: priya@gmail.com
Enter value for phone: 9876543213
Enter value for address: madurai
old 1: INSERT INTO Seller Values ('&Seller_id','&Seller_Name','&email','&Phone','&Address')
new 1: INSERT INTO Seller Values ('S002','Priya','priya@gmail.com',9876543213,'madurai')

1 row created.

SQL> /
Enter value for seller_id: S003
Enter value for seller_name: Karthik
Enter value for email: karthik@gmail.com
Enter value for phone: 9876543214
Enter value for address: Trichy
old 1: INSERT INTO Seller Values ('&Seller_id','&Seller_Name','&email','&Phone','&Address')
new 1: INSERT INTO Seller Values ('S003','Karthik','karthik@gmail.com',9876543214,'Trichy')

1 row created.

SQL> /
Enter value for seller_id: S004
Enter value for seller_name: Meena
Enter value for email: meena@gmail.com
Enter value for phone: 9876543215
Enter value for address: Erode
old 1: INSERT INTO Seller Values ('&Seller_id','&Seller_Name','&email','&Phone','&Address')
new 1: INSERT INTO Seller Values ('S004','Meena','meena@gmail.com',9876543215,'Erode')

1 row created.
```

```
SQL> /
Enter value for seller_id: S005
Enter value for seller_name: Vijay
Enter value for email: vijay@gmail.com
Enter value for phone: 9876543216
Enter value for address: Vellore
old 1: INSERT INTO Seller Values ('&Seller_id','&Seller_Name','&email','&Phone','&Address')
new 1: INSERT INTO Seller Values ('S005','Vijay','vijay@gmail.com','9876543216','Vellore')

1 row created.
```

Insert Values to Inventory Table

```
SQL> INSERT INTO Inventory Values('&Inventory_Id','&Product_Id','&Seller_Id','&Stock_Quantity','&Stock_status', TO_DATE('&Last_Update','YYYY-MM-DD'));
```

```
SQL> /
Enter value for inventory_id: INV005
Enter value for product_id: P04
Enter value for sellr_id: S005
Enter value for stock: 8
Enter value for stock_status: Available
Enter value for last_update: 2026-09-19
old 1: INSERT INTO Inventory VALUES('&Inventory_Id','&Product_Id','&Sellr_Id','&Stock','&Stock_Status',To_DATE('&Last_Update','YYYY-MM-DD'))
new 1: INSERT INTO Inventory VALUES('INV005','P04','S005',8,'Available',To_DATE('2026-09-19','YYYY-MM-DD'))

1 row created.
```

Report 1: Seller-wise Product Report Display:

- Seller name
- Number of products supplied
- Available stock

```
SQL> INSERT INTO Inventory Values('&Inventory_Id','&Product_Id','&Sellr_Id','&Stock','&Stock_Status',To_DATE('&Last_Update','YYYY-MM-DD'));
Enter value for inventory_id: INV001
Enter value for product_id: P01
Enter value for sellr_id: S001
Enter value for stock: 4
Enter value for stock_status: Available
Enter value for last_update: 2026-09-01
old 1: INSERT INTO Inventory VALUES('&Inventory_Id','&Product_Id','&Sellr_Id','&Stock','&Stock_Status',To_DATE('&Last_Update','YYYY-MM-DD'))
new 1: INSERT INTO Inventory VALUES('INV001','P01','S001',4,'Available',To_DATE('2026-09-01','YYYY-MM-DD'))

1 row created.

SQL> /
Enter value for inventory_id: INV002
Enter value for product_id: P02
Enter value for sellr_id: S002
Enter value for stock: 7
Enter value for stock_status: Available
Enter value for last_update: 2026-08-07
old 1: INSERT INTO Inventory VALUES('&Inventory_Id','&Product_Id','&Sellr_Id','&Stock','&Stock_Status',To_DATE('&Last_Update','YYYY-MM-DD'))
new 1: INSERT INTO Inventory VALUES('INV002','P02','S002',7,'Available',To_DATE('2026-08-07','YYYY-MM-DD'))

1 row created.

SQL> /
Enter value for inventory_id: INV003
Enter value for product_id: P03
Enter value for sellr_id: S001
Enter value for stock: 8
Enter value for stock_status: Available
Enter value for last_update: 2026-05-29
old 1: INSERT INTO Inventory VALUES('&Inventory_Id','&Product_Id','&Sellr_Id','&Stock','&Stock_Status',To_DATE('&Last_Update','YYYY-MM-DD'))
new 1: INSERT INTO Inventory VALUES('INV003','P03','S001',8,'Available',To_DATE('2026-05-29','YYYY-MM-DD'))

1 row created.

SQL> /
Enter value for inventory_id: INV004
Enter value for product_id: P04
Enter value for sellr_id: S004
Enter value for stock: 3
Enter value for stock_status: Available
Enter value for last_update: 2026-07-15
old 1: INSERT INTO Inventory VALUES('&Inventory_Id','&Product_Id','&Sellr_Id','&Stock','&Stock_Status',To_DATE('&Last_Update','YYYY-MM-DD'))
new 1: INSERT INTO Inventory VALUES('INV004','P04','S004',3,'Available',To_DATE('2026-07-15','YYYY-MM-DD'))

1 row created.
```

```

SQL> /
Enter value for inventory_id: INV005
Enter value for product_id: P04
Enter value for sellr_id: S005
Enter value for stock: 8
Enter value for stock_status: Available
Enter value for last_update: 2026-09-19
old 1: INSERT INTO Inventory VALUES('&Inventory_Id','&Product_Id','&Sellr_Id','&Stock','&Stock_Status',To_DATE('&Last_Update','YYYY-MM-DD'))
new 1: INSERT INTO Inventory VALUES('INV005','P04','S005',8,'Available',To_DATE('2026-09-19','YYYY-MM-DD'))

1 row created.

```

```

SQL> SELECT s.Seller_Name,
2         COUNT(i.Product_Id) AS Number_of_Products,
3         SUM(i.Stock_Quantity) AS Available_Stock
4 FROM Seller s
5 JOIN Inventory i
6 ON s.Sellr_Id = i.Seller_Id
7 WHERE i.Stock_Status = 'Available'
8 GROUP BY s.Seller_Name;

```

SELLER_NAME	NUMBER_OF_PRODUCTS	AVAILABLE_STOCK
Meena	1	3
Arjun	2	12
Priya	1	7
Vijay	1	8

```

SQL> SELECT * FROM Seller;

```

SELLR_I	SELLER_NAME	EMAIL	PHONE
ADDRESS			
S001	Arjun	arjun@gmail.com	9876543211
Salem			
S002	Priya	priya@gmail.com	987654312
Madurai			
S003	Karthik	karthik@gmail.com	9876543213
Trichy			
SELLR_I	SELLER_NAME	EMAIL	PHONE
ADDRESS			
S004	Meena	meena@gmail.com	9876543214
Erode			
S005	Vijay	vijay@gmail.com	9876543215
Vellore			

Use Joins to get details from Seller and Inventory Table.

Report 2: Stock Availability Report Display:

- Product name
- Stock quantity
- Stock status

```
SQL> SELECT p.Product_Name, i.Stock_Quantity, i.Stock_Status FROM Product p JOIN Inventory i on p.product_Id = i.Product_Id;
```

PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATU
Mixer Grinder	4	Available
Electric kettle	7	Available
Face cream	8	Available

PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATU
Basmathi rice	3	Available
Basmathi rice	8	Available

Report 3: Inventory Analysis Find:

- Total products available

```
SQL> SELECT COUNT(*) AS Total_Available_Product FROM Inventory WHERE Stock_Status = 'Available';
```

TOTAL_AVAILABLE_PRODUCT
5

- Products out of stock

```
SQL> SELECT p.Product_Name, i.Stock_Quantity FROM Product p JOIN Inventory i
ON p.Product_Id = i.Product_Id WHERE i.Stock_Quantity = (SELECT MAX(Stock_Q
uantity) FROM Inventory);
```

```
PRODUCT_NAME
```

```
-----
```

```
STOCK_QUANTITY
```

```
-----
```

```
Face cream
```

```
8
```

```
Basmathi rice
```

```
8
```

- Highest stocked products

```
SQL> SELECT AVG(Stock_Quantity) AS Average_Inventory FROM Inventory;
```

```
AVERAGE_INVENTORY
```

```
-----
```

```
6
```

- Average inventory quantity

```
SQL> SELECT COUNT(*) AS Out_of_Stock FROM Inventory WHERE Stock_Status = 'Ou
t of Stock';
```

```
OUT_OF_STOCK
```

```
-----
```

```
0
```